



Plan to Empower.

FINAL REPORT

Brecksville-Broadview Heights City Schools
Enrollment Projections
January 15, 2013

Brecksville-Broadview Heights City Schools, OH

INTRODUCTION

This report contains ten-year enrollment projections for the Brecksville-Broadview Heights City School District. Enrollment projections were developed by analyzing the following data:

- ▶ Live birth data
- ▶ Historical enrollment
- ▶ Non-public school enrollment
- ▶ Community demographics
- ▶ Housing information

The projections presented in this report are meant to serve as a planning tool for the future, and, with the historical data and anticipated growth, represent the most likely direction of the District.

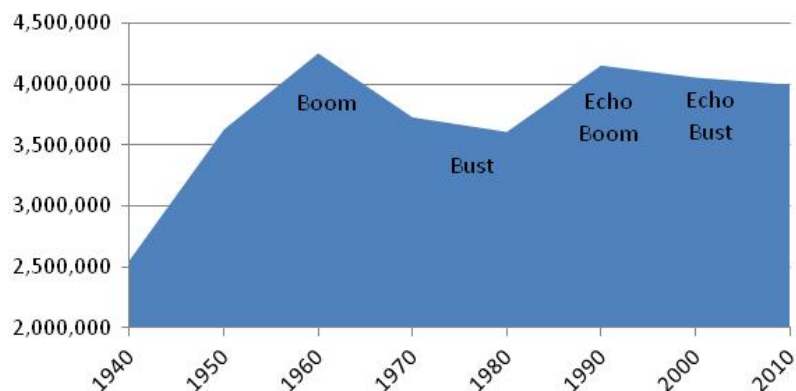
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ENROLLMENT PROJECTION METHODOLOGIES

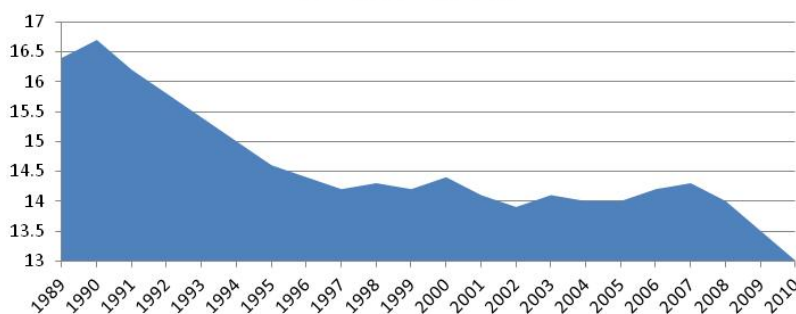
Introduction

Tracing the landscape of the country's public school enrollment back over the past fifty years reveals demographic, economic, and social changes. The United States as a whole continues to undergo major shifts in public student enrollment, due in large part to past events including the baby boom, the availability and use of birth control, and the development of suburbs. The baby boom of the late 1940s and 50s was followed by the baby bust of the 1960s and 70s. This gave rise to the echo baby boom of the 1980s, producing children who primarily graduated in the late 90s through 2007.

US Total Live Births



US Population - Live Birth Rate per 1,000 Population



increased again in 2000. However, the birth rate resumed a descending pattern in 2001 and reached an all-time low of 13.0 in 2010.

Nationwide, districts continue to experience the effects of the echo baby bust of the 1990s. From the 1950s to the 1970s, a dramatic downsizing of the family unit occurred. A direct result was the declining school enrollment of the 1970s and 1980s. As of the 2010 Census, the size of a family was at an all-time low of 3.14 persons. The live birth rate increased for the first time in several years in 1998 and

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When projecting future enrollments, it will be vital to track the number of live births, the amount of new housing activity, and the change in household composition.

In addition, any of the following factors could cause a significant change in projected student enrollments:

- ▶ Boundary adjustments
- ▶ New school openings
- ▶ Changes/additions in program offerings
- ▶ Preschool programs
- ▶ Change in grade configuration
- ▶ Interest rates/unemployment shifts
- ▶ Magnet/Charter/Private school opening or closure
- ▶ Zoning changes
- ▶ Unplanned new housing activity
- ▶ Planned, but not built, housing

Obviously, certain factors can be gauged and planned for far better than others. For instance, it may be relatively straightforward to gather housing data from local builders regarding the total number of lots in a planned subdivision and calculate the potential student yield. However, planning for changes in the unemployment rate, and how these may either boost or reduce public school enrollment, proves more difficult. In any case, it is essential to gather a wide variety of information in preparation for producing enrollment projections.

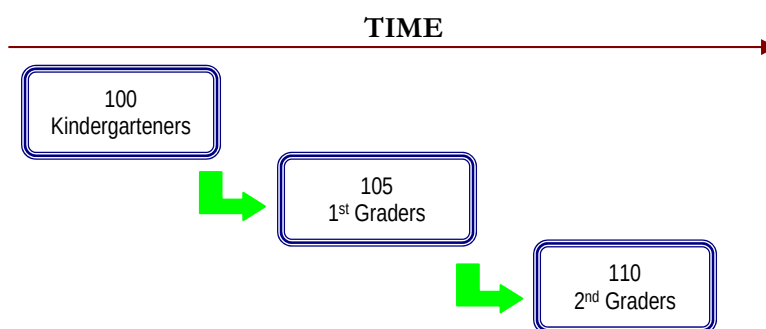
When looking ahead at a school district's enrollment over the next two, five, or ten years, it is helpful to approach the process from a global perspective. For example: How many new homes have been constructed each year? How many births have occurred each year in relation to the resident population? Is housing experiencing a turnover – if so, what is the composition of families moving in/out? Are more or less students attending private school or being home-schooled? What has the unemployment rate trend been over the past ten years? What new educational policies are in place now that could affect student enrollment figures?

The data sets generated from questions such as these have led to the development of general methodologies to project future student enrollments. They are as follows:

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Cohort Survival Method

A cohort is a group of persons [in this case, students]. The cohort survival projection methodology uses previous live birth data and historical student enrollments to “age” a known population or cohort throughout the school grades. For instance, a cohort begins when a group of kindergarteners enrolls in grade K and moves to first grade the following year, second grade the next year, and so on.



A “survival ratio” is developed to track how this group of students grew or shrunk in number as they moved through the grade levels. By determining survival ratios for each grade transition [i.e., 1st to 2nd grade] over a ten-year period of time, patterns emerge and projection ratios can be developed to be used as a multiplier.

For example, if student enrollment has consistently increased from the 8th to the 9th grade over the past ten years, the survival ratios for each year would be greater than 100 percent. Through analysis of the survival ratios, the projection ratio is determined and is multiplied by the current 8th grade to develop a projection for next year’s 9th grade.

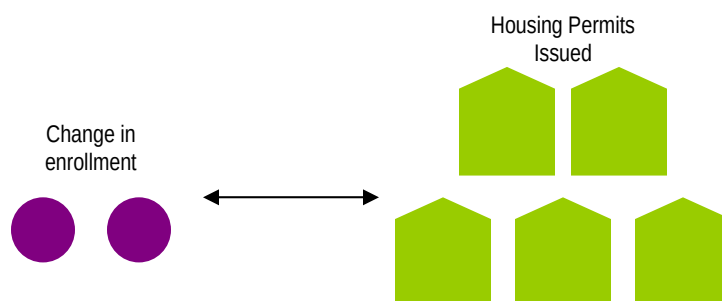
This methodology can be carried through to develop ten years of projection figures. Because there is not a grade cohort to follow for students coming into kindergarten, live birth counts are used to develop a survival ratio. Babies born five years previous to the kindergarten class are compared in number, and a ratio can be developed to project future kindergarten enrollments.

The cohort survival method is useful in areas where population is stable [relatively flat, growing steadily, or declining steadily], and where there have been no significant fluctuations in enrollment, births, and housing patterns from year to year.

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Housing

Enrollment projections can also be determined by analyzing the housing data for the areas that make up a school district. Yield factors can be established by comparing the historic change in enrollment from year to year divided by the total number of building or occupancy permits issued. For example, if student enrollment has increased by approximately 100 students each year and approximately 200 building permits have been issued each year for the past ten years, then the yield factor would be approximately .5 students per building permit.



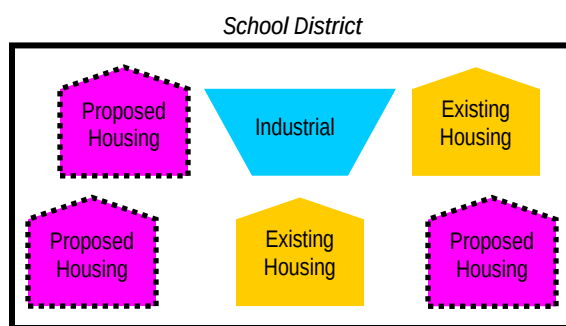
Once yield factors are established, the number of new students per year can be estimated by multiplying the yield factor by the number of projected new housing units. This method is effective when the rate of kindergarten enrollment far exceeds the live birth counts.

If housing demolitions are occurring in a district, these must also be taken into account. For instance, if housing demolitions/withdrawals have increased rapidly over recent years while new housing starts have remained relatively constant over many years, the conclusion may be that some of the new housing starts will simply be replacements for the families displaced by the demolitions. Of course, housing value and household composition would need to be further analyzed to confirm that this is indeed the case. It is possible that enrollment may remain flat or decline even though there is new housing occurring in the area.

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Land-Saturation Analysis

Housing data also drives the land-saturation analysis enrollment methodology. In areas where there is a high rate of development and the future development patterns in the area are clear, a “build-out” scenario can be developed. The scenario takes into consideration the remaining acreage to be developed, planned rate of completion, zoning policies, density per acre, type of housing, and ratios of school-age children per household type. This method is particularly useful in areas experiencing rapid growth.



Geographic Information Systems

While not a methodology, the need for better tools and easier manipulation of data has led to a new industry standard in planning – GIS [Geographic Information Systems]. GIS technology allows school districts to quickly analyze countless data sets including birth data, housing information, and enrollment statistics.

When paired with enrollment projections, GIS becomes an invaluable information-management and decision-making tool. Often, county or city offices are already implementing GIS technology and data can be shared and expanded among these organizations in the district. GIS tables and maps are included within this report illustrating population, age, and income estimates and projections.

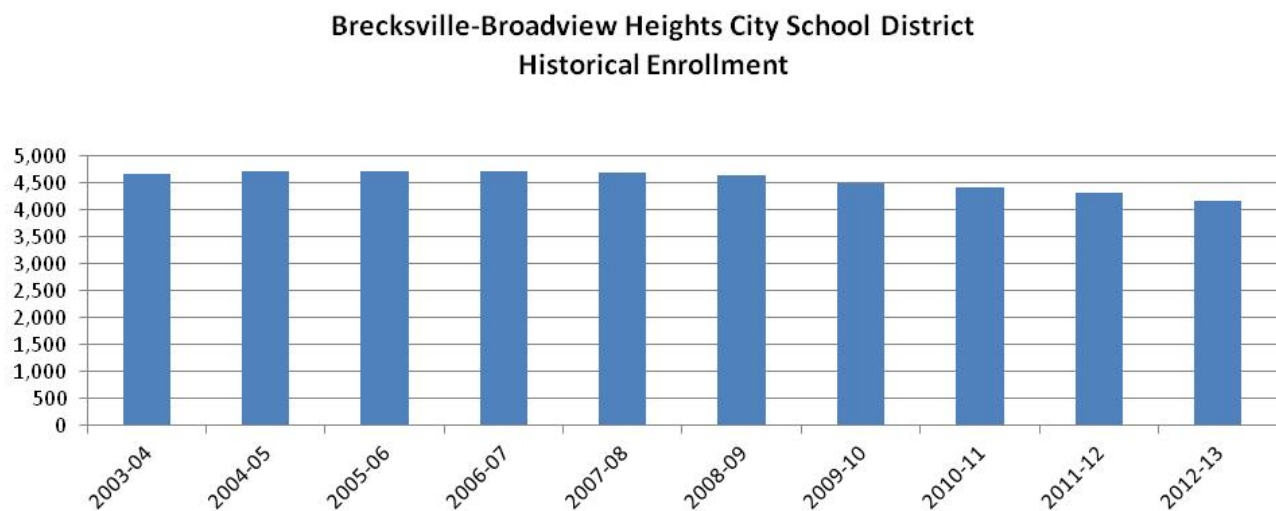
The cohort survival was the primary method used in the development of the enrollment projections for the Brecksville-Broadview Heights City School District.

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HISTORICAL ENROLLMENT

Over the past ten years, student enrollment in the Brecksville-Broadview Heights City School District has decreased by 493 students in grades Pre-K – 12, including ungraded students. Total enrollment for the 2012-13 school year is 4,173 students.

The following graph illustrates the District's Pre-K – 12 enrollment history from 2003-04 through 2012-13.



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The following table illustrates the District's enrollment history from 2003-04 through 2012-13.

**Brecksville-Broadview Heights City School District
Historical Enrollment**

Grade	2003-04	2004-05	2005-06	2006-07	2007-08	2008-09	2009-10	2010-11	2011-12	2012-13
Pre-K	36	39	44	48	49	67	54	63	57	64
K	285	250	228	228	229	261	250	222	220	206
1	301	321	299	299	311	251	269	266	233	245
2	318	321	324	324	270	326	256	284	278	238
3	364	329	332	332	332	271	335	274	291	289
4	353	372	340	340	356	341	282	346	283	303
5	380	366	377	377	341	367	343	286	350	287
6	339	386	383	383	357	342	370	357	309	350
7	414	369	400	400	397	371	347	366	362	315
8	385	409	377	377	405	406	370	348	369	370
9	380	411	430	430	409	417	415	366	367	364
10	392	376	407	407	402	418	413	414	356	365
11	362	397	372	372	433	389	418	414	413	361
12	339	380	407	407	406	423	377	413	417	408
Ungraded	18	2	0	0	1	4	9	10	10	8
Total	4,666	4,728	4,720	4,724	4,698	4,654	4,508	4,429	4,315	4,173

Source: Brecksville-Broadview Heights City School District

**Brecksville-Broadview Heights City School District
Historical Enrollment by Grade Group**

Grade	2003-04	2004-05	2005-06	2006-07	2007-08	2008-09	2009-10	2010-11	2011-12	2012-13
Pre-K - 3	1,304	1,260	1,227	1,231	1,191	1,176	1,164	1,109	1,079	1,042
4 - 5	733	738	717	717	697	708	625	632	633	590
6 - 8	1,138	1,164	1,160	1,160	1,159	1,119	1,087	1,071	1,040	1,035
9 - 12	1,473	1,564	1,616	1,616	1,650	1,647	1,623	1,607	1,553	1,498
Ungraded	18	2	0	0	1	4	9	10	10	8
Total	4,666	4,728	4,720	4,724	4,698	4,654	4,508	4,429	4,315	4,173

Source: Brecksville-Broadview Heights City School District

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LIVE BIRTH DATA

Utilization of live birth data is recommended when projecting future kindergarten enrollments as it provides a helpful overall trend. The live birth counts are used in determining a birth-to-kindergarten survival ratio. This ratio identifies the percentage of children born in a representative area who attend kindergarten in the District five years later. The survival ratios for birth-to-kindergarten as well as grades 1-12 can be found later in this report.

The Ohio Department of Health [ODH] data warehouse provides information about live birth events for Ohio residents. Information about events occurring outside of Ohio to Ohio residents is included. Information about events occurring inside Ohio to non-Ohio residents is not included.

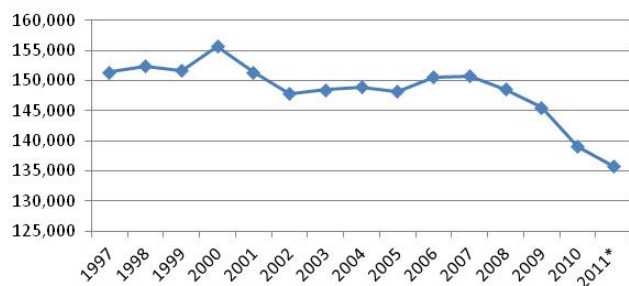
Data is arranged by the residence of the mother. For example, if a mother lives in Powell, Delaware County but delivers her baby in Columbus, Franklin County, the birth is counted in Powell, Delaware County.

The number of live births is recorded by:

- ▶ State
- ▶ County
- ▶ City/Town
- ▶ Census Tract
- ▶ Zip Code
- ▶ Address [not available to the public]

Live birth counts are different from live birth rates. The live birth count is the actual number of live births. A birth rate is the number of births per 1,000 women in a specified population group. Birth rates are provided for counties only and for 9 age groups from 10-14 years to 45+ years.

Ohio Live Birth Counts



Ohio has experienced the same trend in live births as seen around the country. Live birth counts increased for the first time in several years in 1998 and again in 2000. A descending pattern resumed in 2001 with a slight stabilization from 2002 to 2005. Births increased again in 2006 and 2007 but have since declined to an all-time low of 135,815 in 2011.

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The following chart and graph include the live birth count for zip codes 44141 and 44147. The peak year for births was 1999 at 403 while 2010 had the lowest number of births with 342.

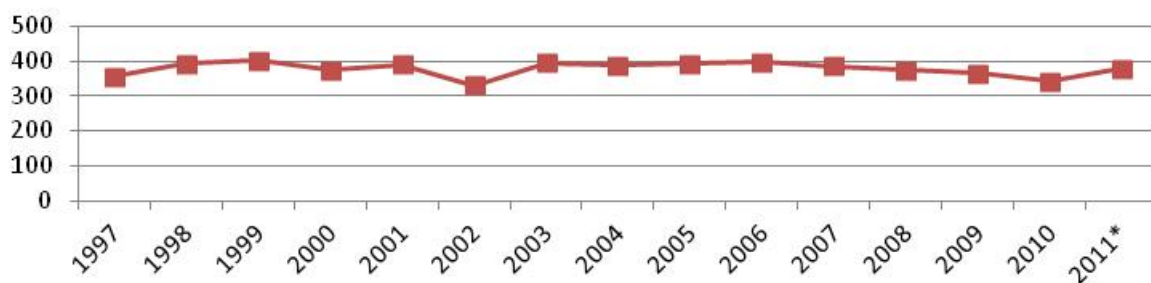
**Brecksville-Broadview Heights City School District
Birth Count by Zip Code
1997-2011***

Year	44141	44147	Total
1997	197	160	357
1998	216	177	393
1999	215	188	403
2000	194	179	373
2001	192	199	391
2002	174	157	331
2003	193	202	395
2004	188	199	387
2005	194	198	392
2006	173	224	397
2007	154	232	386
2008	164	210	374
2009	160	205	365
2010	169	173	342
2011*	174	205	379

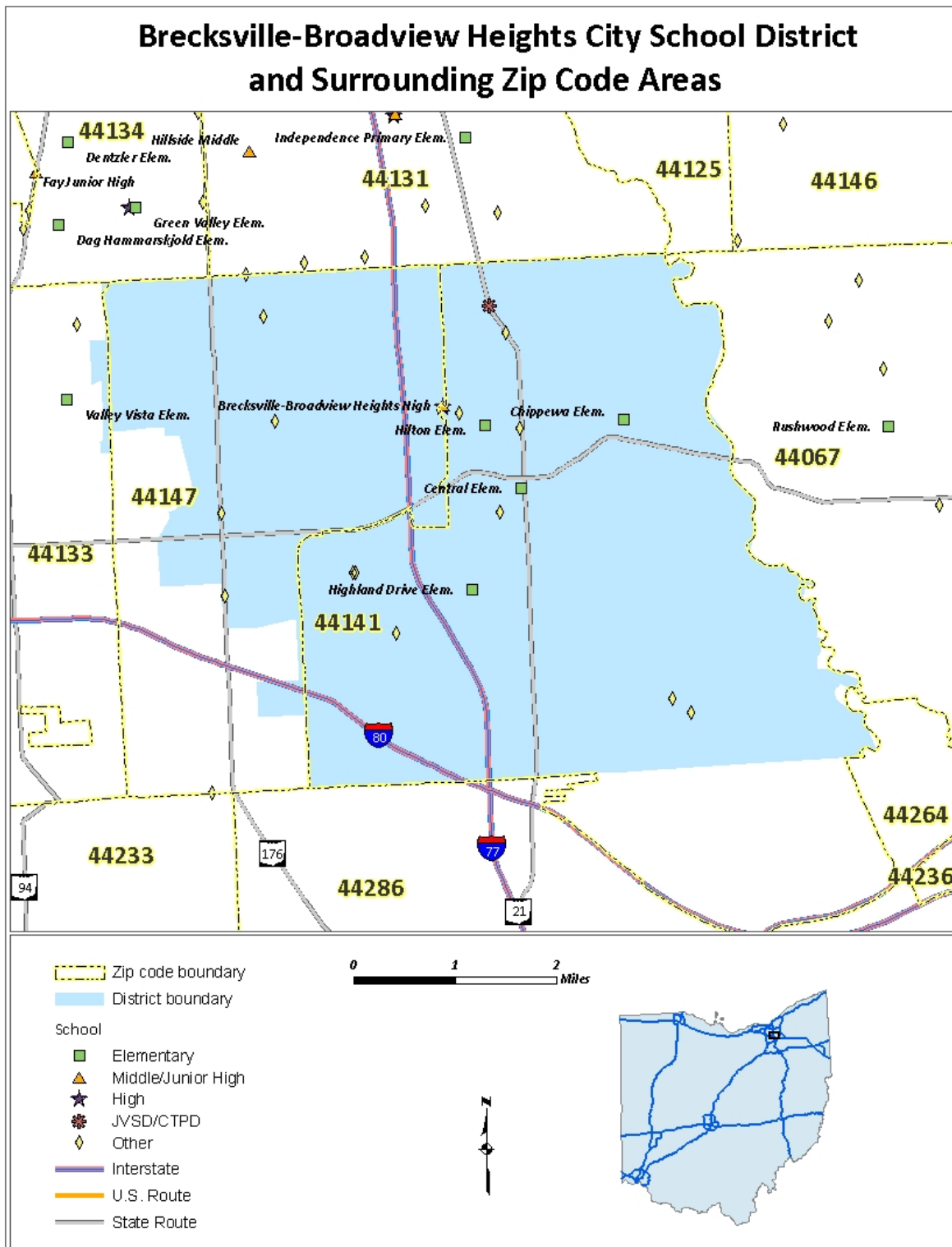
Source: Ohio Department of Health, Statistical Analysis Unit

* provisional

**Brecksville-Broadview Heights City School District
Birth Count by Zip Code**



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DEMOGRAPHICS

The Brecksville-Broadview Heights City School District is comprised of Brecksville City, Broadview Heights City, and North Royalton City in Cuyahoga County. General demographic data is included in the following tables for the areas located in the District.

General Demographic Information

	Cuyahoga County
Per Capita Income	\$25,796
Median Household Income	\$41,530
Persons Below Poverty	18.6%

Source: US Census, American Community Survey 2011 1-Year Estimates

Total Population

	2000 Census	2010 Census
Cuyahoga County	1,393,978	1,280,122
Brecksville city	13,382	13,656
Broadview Heights city	15,967	19,400
North Royalton city	28,648	30,444

Source: ODOD Policy Research & Strategic Planning Office, August 2011

Also included are block group estimates and projections provided by ESRI Business Information Solutions (ESRI BIS). ESRI BIS uses a time series of estimates from the U.S. Census Bureau that includes the latest estimates and intercensal estimates adjusted for error of closure. The Census Bureau's time series is consistent, but testing has revealed improved accuracy by using a variety of sources to track county population trends.

ESRI BIS also employs a time series of building permits and housing starts plus residential deliveries. Finally, local data sources that tested well against Census 2000 are reviewed. Data sources are integrated and then analyzed by Census Block Groups.

Sources of data include:

- ▶ Supplementary Surveys of the Census Bureau
- ▶ Bureau of Labor Statistics' (BLS) Local Area Unemployment Statistics
- ▶ BLS Occupational Employment Statistics
- ▶ InfoUSA
- ▶ U.S. Bureau of the Census' Current Population Survey
- ▶ National Planning Association Data Service

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Below is a list of definitions as they appear on the U.S. Census Bureau website, to aid in interpretation of the following tables and maps.

Household:

A household includes all the people who occupy a housing unit as their usual place of residence.

Average family size:

A measure obtained by dividing the number of members of families by the total number of families (or family householders).

Family household (Family):

A family includes a householder and one or more people living in the same household who are related to the householder by birth, marriage, or adoption. All people who are related to the householder are regarded as members of his or her family. A family household may contain people not related to the householder, but those people are not included as part of the householder's family in census tabulations. Thus, the number of family households is equal to the number of families, but family households may include more members than do families. A household can contain only one family for purposes of census tabulations. Not all households contain families since a household may comprise a group of unrelated people or one person living alone.

Householder:

The person, or one of the people, in whose name the home is owned, being bought, or rented. If there is no such person present, any household member 15 years old and over can serve as the householder for the purposes of the census. Two types of householders are distinguished: a family householder and a nonfamily householder. A family householder is a householder living with one or more people related to him or her by birth, marriage, or adoption. The householder and all people in the household related to him are family members. A nonfamily householder is a householder living alone or with nonrelatives only.

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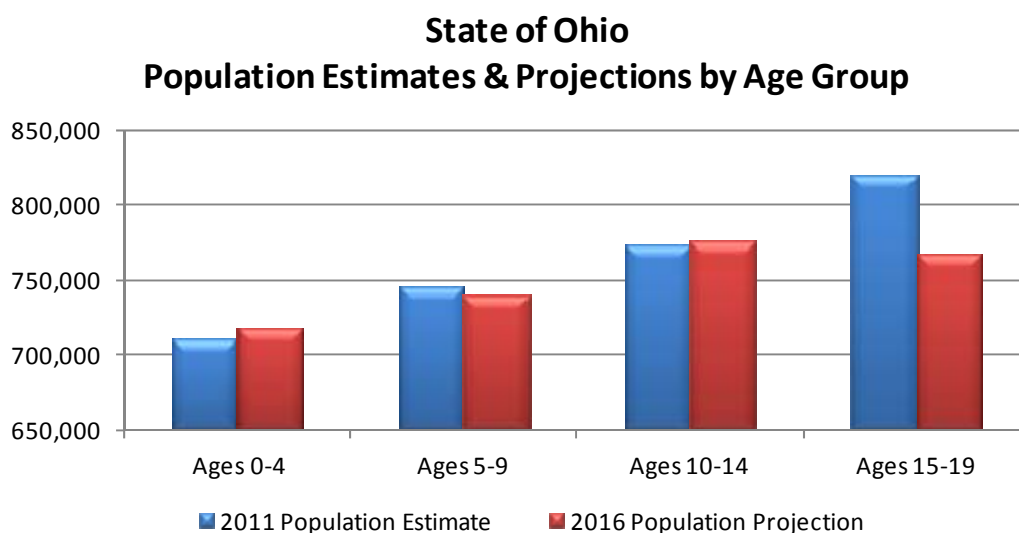
The following tables illustrate the current estimates and 5-year population projections based on block groups that comprise the state and school district, indicating areas of current and projected growth. The tables have been developed to determine selected age group projections and projections for household income, family size, and family income.

The total population in the State of Ohio is 11,538,102. This population is projected to increase by 57,804 people, or approximately 0.5% over a 5-year period. However, Ohio gained just 8,400 residents between 2010 and 2011, ranking the state 48th for growth, according to new population estimates from the U.S. Census.

The 0-19 year-old population in the State currently totals 3,047,034. This population is projected to decrease by 48,194 children, or approximately 2 percent.

State of Ohio	2011 Population Estimate	2016 Population Projection
Total Population	11,538,102	11,595,906
Ages 0-4	710,393	717,415
Ages 5-9	744,959	739,354
Ages 10-14	772,909	775,032
Ages 15-19	818,772	767,040
Total Ages 0-19	3,047,034	2,998,840

Source: ESRI BIS



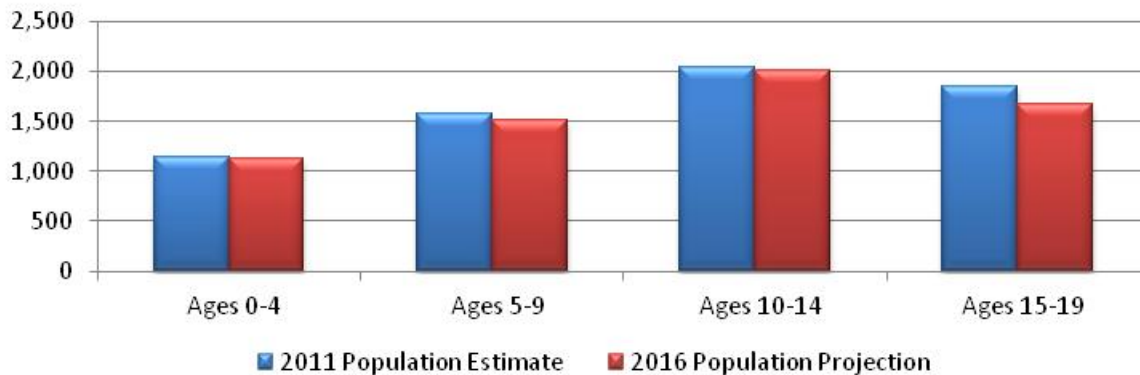
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The total population in the District currently totals 26,065. This population is projected to decrease by 611 children, or approximately 2 percent. The 0-19 year-old population in the District currently totals 6,607. This population is projected to decrease by 299 children, or approximately 5 percent.

Brecksville-Broadview Heights City School District	2011 Population Estimate	2016 Population Projection
Total Population	26,065	25,454
Ages 0-4	1,142	1,120
Ages 5-9	1,574	1,515
Ages 10-14	2,045	1,999
Ages 15-19	1,846	1,674
Total Ages 0-19	6,607	6,308

Source: ESRI BIS

**Brecksville-Broadview Heights City School District
Population Estimates & Projections by Age Group**



Brecksville-Broadview Heights City Schools, OH

Average household and family incomes in the State are projected to increase by 11% and 12%, respectively over a 5-year period. Average family size is projected to remain relatively the same.

State of Ohio	2011 Population Estimate	2016 Population Projection
Average Household Income	\$61,196	\$67,985
Average Family Size	3.01	3.00
Average Family Income	\$94,080	\$105,718

Source: ESRI BIS

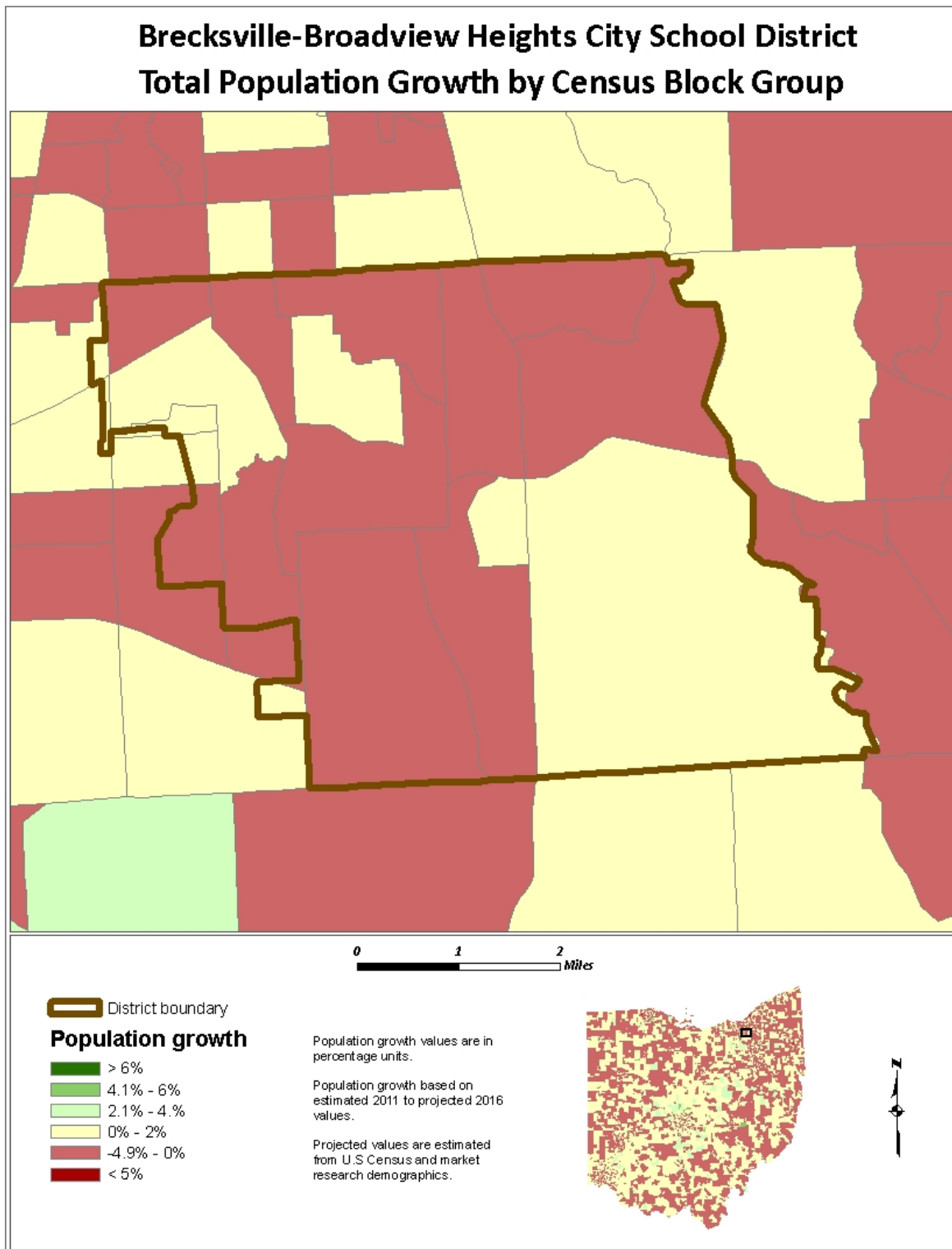
Average household and family incomes in the District are projected to increase by 10% and 11%, respectively over a 5-year period. Average family size is projected to remain relatively the same.

Brecksville-Broadview Heights City School District	2011 Population Estimate	2016 Population Projection
Average Household Income	\$94,715	\$104,532
Average Family Size	3.06	3.05
Average Family Income	\$132,437	\$147,178

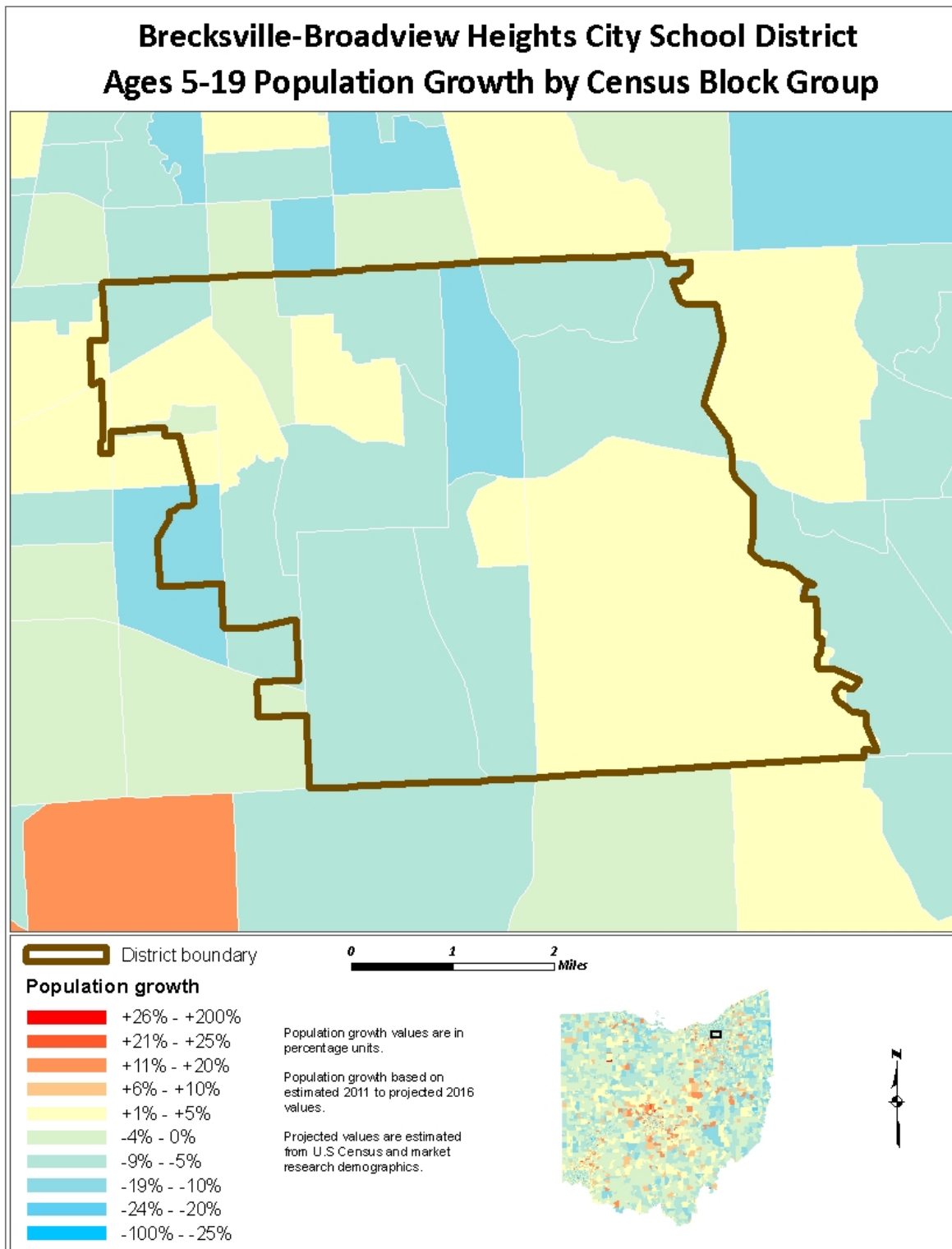
Source: ESRI BIS

The maps on the following pages illustrate the data identified in the tables. The color coding identifies areas within the District that may be increasing or decreasing at different rates than others.

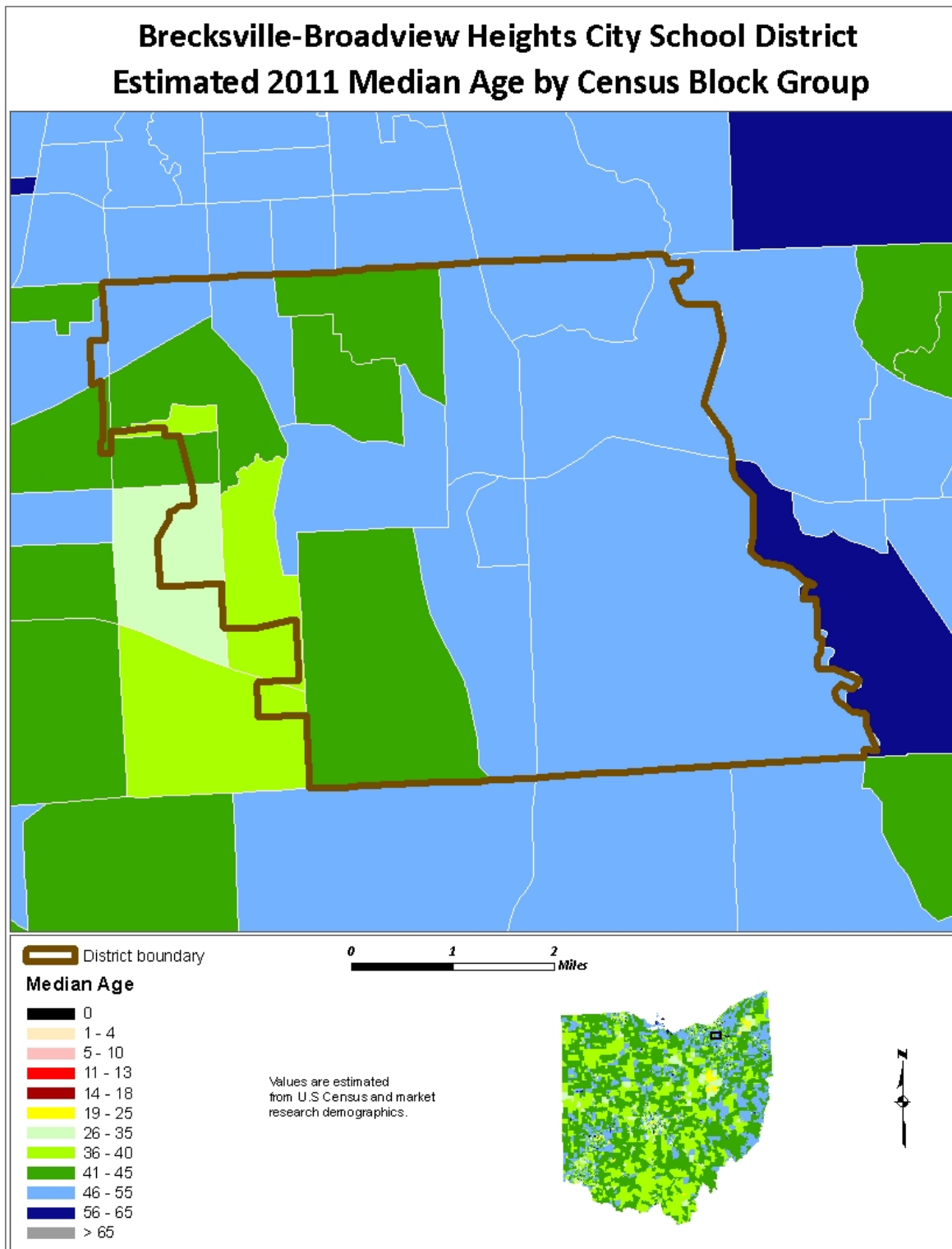
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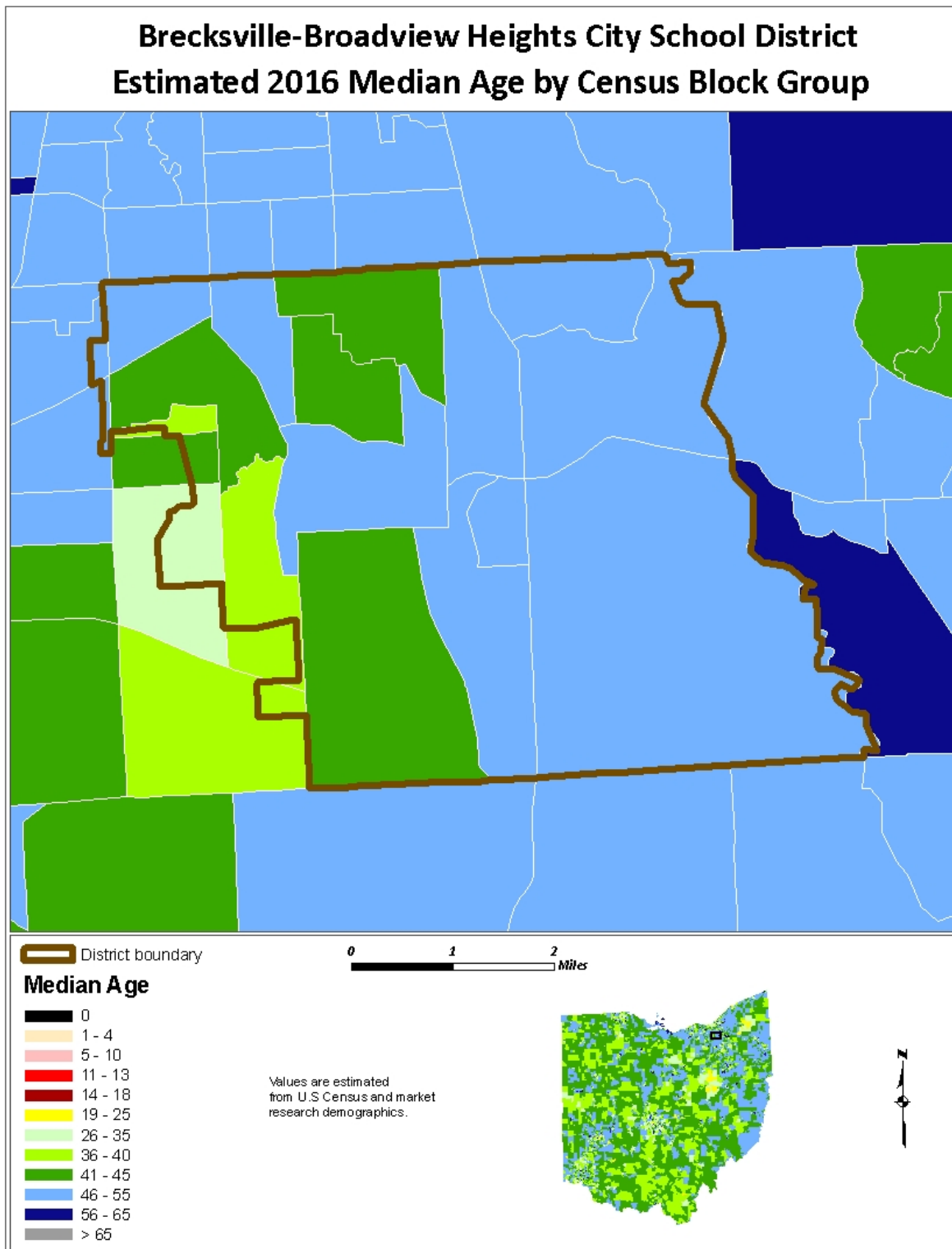
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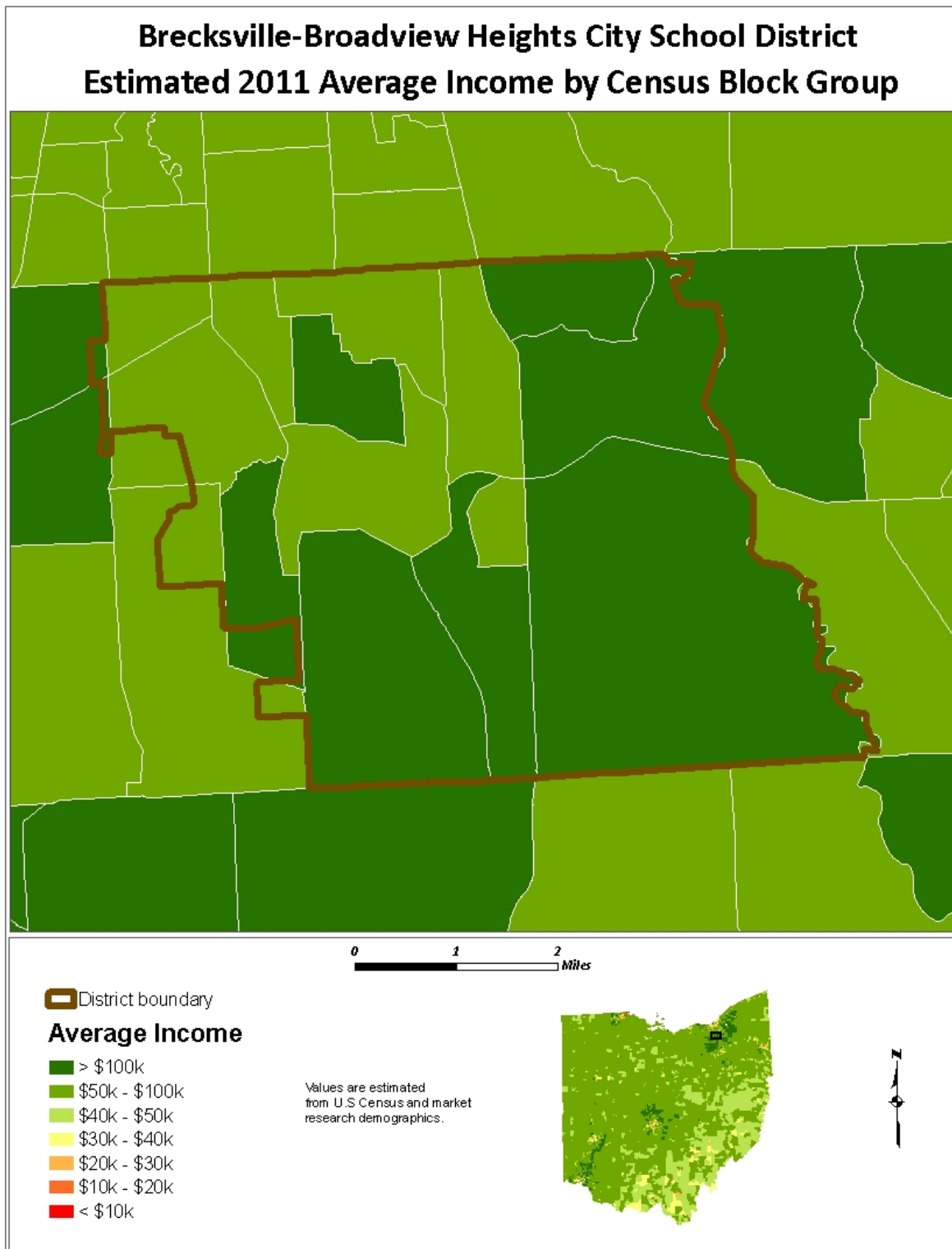
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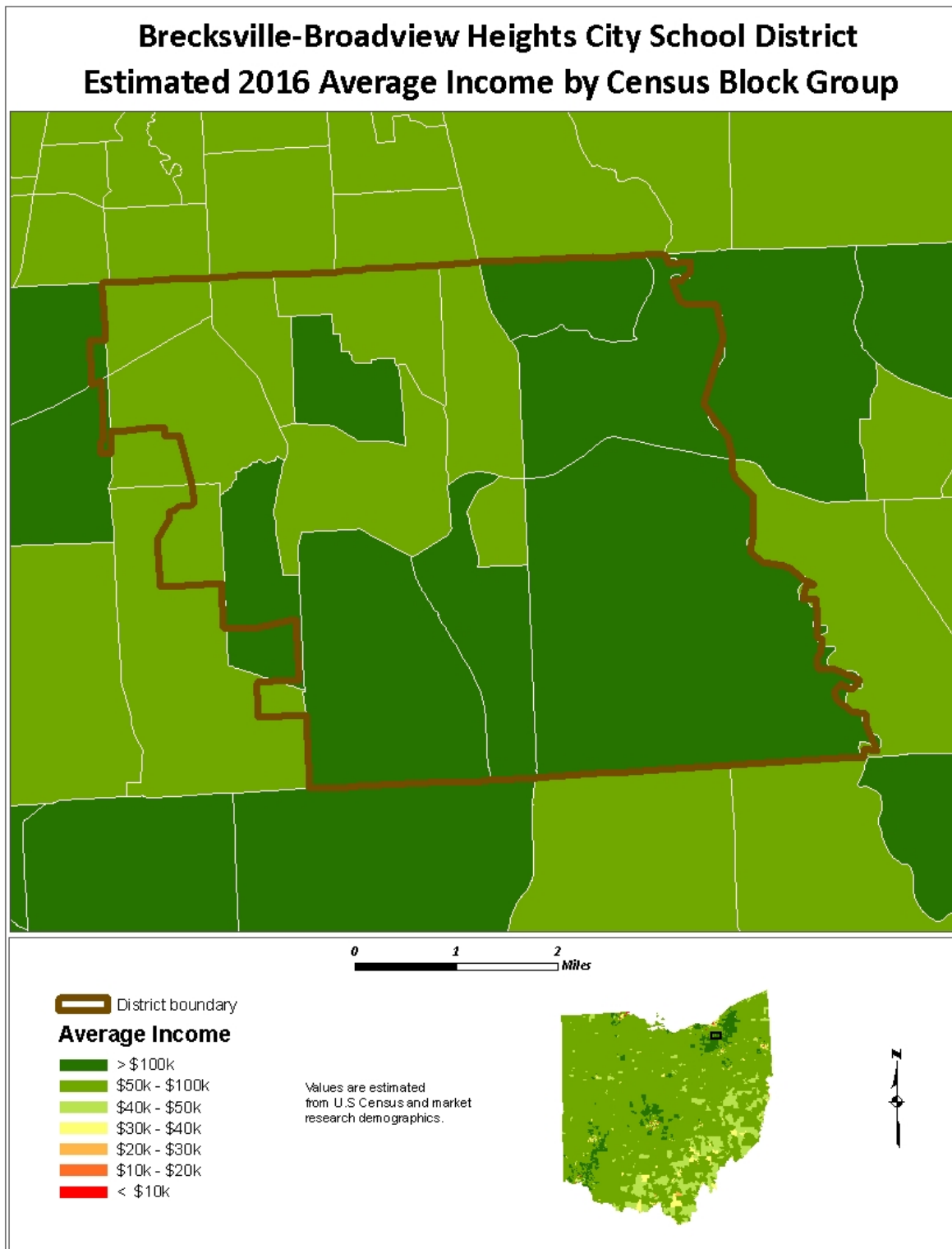
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HOUSING INFORMATION

The chart below illustrates the number of single-family dwelling building permits issued each year in Brecksville, Broadview Heights, and Cuyahoga County. The number of permits issued has dropped dramatically since 2004 (the peak year for both Broadview Heights and Cuyahoga County).

of Building Permits Issued for Single Family Dwellings

Year	Brecksville	Broadview Heights	Cuyahoga County
2001	65	104	1,810
2002	37	258	1,805
2003	46	272	1,865
2004	29	342	2,010
2005	40	177	1,748
2006	23	50	1,246
2007	12	30	922
2008	12	18	542
2009	9	12	462
2010	4	23	490
2011	4	21	401
2012*	6	20	352

Source: SOCDS Building Permits Database

* through September 2012

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SURVIVAL RATIOS

The chart below demonstrates the changes in enrollment as students move through the system. Percentages greater than 100 indicate that there are more students than there were in the previous grade the previous year. In other words, there was growth and new students entered the system. Percentages less than 100 indicate that there was decline with students leaving the system.

- Birth to Kindergarten: This ratio indicates the number of children born in the area who attend kindergarten in the District 5 years later. Percentages less than 100% result from movement out of the district, attendance at a non-public or charter school, or residence in another district within the same area.

The following table illustrates the survival ratios used for the Brecksville-Broadview Heights City School District.

from	to	birth -> K	K->1	1->2	2->3	3->4	4->5	5->6	6->7	7->8	8->9	9->10	10->11	11->12
2003	2004	62.0%	112.6%	106.6%	103.5%	102.2%	103.7%	101.6%	108.8%	98.8%	106.8%	98.9%	101.3%	105.0%
2004	2005	61.1%	119.6%	100.9%	103.4%	103.3%	101.3%	104.6%	103.6%	102.2%	105.1%	99.0%	98.9%	102.5%
2005	2006	58.3%	131.1%	108.4%	102.5%	102.4%	110.9%	101.6%	104.4%	94.3%	114.1%	94.7%	91.4%	109.4%
2006	2007	69.2%	136.4%	90.3%	102.5%	107.2%	100.3%	94.7%	103.7%	101.3%	108.5%	93.5%	106.4%	109.1%
2007	2008	66.1%	109.6%	104.8%	100.4%	102.7%	103.1%	100.3%	103.9%	102.3%	103.0%	102.2%	96.8%	97.7%
2008	2009	64.6%	103.1%	102.0%	102.8%	104.1%	100.6%	100.8%	101.5%	99.7%	102.2%	99.0%	100.0%	96.9%
2009	2010	56.6%	106.4%	105.6%	107.0%	103.3%	101.4%	104.1%	98.9%	100.3%	98.9%	99.8%	100.2%	98.8%
2010	2011	55.4%	105.0%	104.5%	102.5%	103.3%	101.2%	108.0%	101.4%	100.8%	105.5%	97.3%	99.8%	100.7%
2011	2012	53.4%	111.4%	102.1%	104.0%	104.1%	101.4%	100.0%	101.9%	102.2%	98.6%	99.5%	101.4%	98.8%
	average	60.75%	115.018%	102.81%	103.2%	103.63%	102.7%	101.7%	103.1%	100.2%	104.738%	98.204%	99.575%	102.107%
	standard deviation	4.966%	11.081%	4.958%	1.672%	1.418%	3.088%	3.485%	2.593%	2.387%	4.548%	2.530%	3.778%	4.490%

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ENROLLMENT PROJECTION

Enrollment projections were developed after analyzing the data collected in this report. The projections indicate a decrease of 605 students in grades Pre-K through 12, including ungraded students, from the 2012-13 to the 2022-23 school year. The following tables and graph illustrate projected enrollments by grade and by grade group through the 2022-23 school year.

Brecksville-Broadview Heights City School District
Projected Enrollment

Grade	2012-13 Actual	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23
Pre-K	64	69	54	51	56	55	55	54	54	54	54
K	206	223	217	204	226	220	220	219	219	218	218
1	245	228	247	241	225	250	244	244	243	243	242
2	238	254	237	256	250	234	259	253	253	252	252
3	289	248	265	247	266	260	244	270	264	263	263
4	303	300	257	275	256	276	270	253	280	273	273
5	287	307	304	260	278	259	280	273	256	284	277
6	350	296	317	313	269	287	267	289	282	264	293
7	315	353	299	320	316	271	290	270	291	284	267
8	370	318	356	302	323	319	274	292	272	294	287
9	364	375	322	361	306	327	323	277	296	276	298
10	365	360	371	319	357	302	323	320	274	293	273
11	361	365	360	371	319	357	302	323	320	274	293
12	408	357	361	356	366	315	353	299	319	316	271
Ungraded	8	8	8	7	7	7	7	7	7	7	7
Total	4,173	4,061	3,975	3,883	3,820	3,739	3,711	3,643	3,630	3,595	3,568

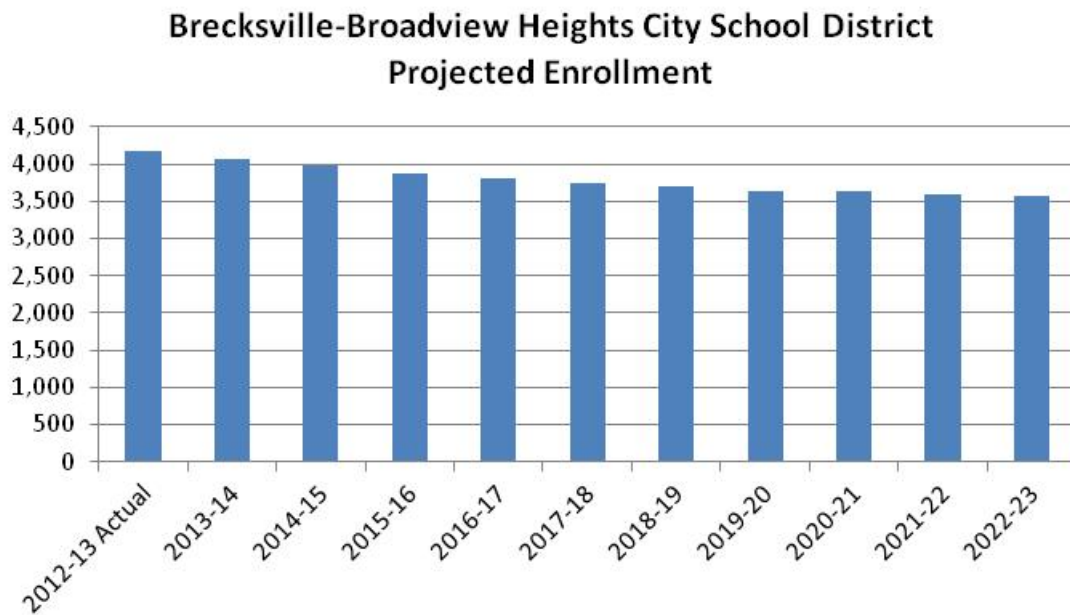
Source: DeJONG-HEALY

Brecksville-Broadview Heights City School District
Projected Enrollment by Grade Group

Grade	2012-13 Actual	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23
Pre-K - 3	1,042	1,022	1,020	999	1,023	1,019	1,022	1,040	1,033	1,030	1,029
4 - 5	590	607	561	535	534	535	550	526	536	557	550
6 - 8	1,035	967	972	935	908	877	831	851	845	842	847
9 - 12	1,498	1,457	1,414	1,407	1,348	1,301	1,301	1,219	1,209	1,159	1,135
Ungraded	8	8	8	7	7	7	7	7	7	7	7
Total	4,173	4,061	3,975	3,883	3,820	3,739	3,711	3,643	3,630	3,595	3,568

Source: DeJONG-HEALY

Brecksville-Broadview Heights City Schools, OH



Brecksville-Broadview Heights City Schools, OH

CONCLUSION

As with any projection, the District should pay close attention to live birth counts, enrollment in elementary schools, community school enrollment, open enrollment, and any housing growth. Each of these factors will have an impact on future student enrollment.

DeJONG-HEALY is pleased to have had the opportunity to provide the District with enrollment projection services. We hope this document will provide the necessary information to make informed decisions about the future of the Brecksville-Broadview Heights City School District.